engineering 4 us all

S Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element K Initial Template - AT - 9 points

Element K: Reflection on the Design Project

Engineering Design Process Summary *(K1, K2)*

Summarize each step of the design process, including:

- Defining the problem (elements A-C)

All the elements were easily completed, everything was or at least felt, written with enough understanding of the problem and knowledge to carry us through this project. Due to the fact that the team was rushing this bit of the project, due to our lack of interest in paperwork, I don't think that anyone really understood the breadth and depth of the situation, leaving us with only the idea that we knew what we were doing, when we had only scratched a couple layers.

- Ideation (element D)

The overall process of the assignment was easy, most of the team participated and shared ideas, making it go much faster. However, I wish that we had spent more time considering our options rather than haphazardly jumping to the design that seemed the simplest as fast as we could, as the lack of future planning ended up causing problems.

- STEM and Justification of design (elements E-F)

I did not work directly on E so I do not have any reflection for it. In element F, nearly everyone participated and did some amount of substantial work and aside from some confusions with the instructions there wasn't anything that went badly.

- Prototype building (element G):

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During this stage of development everything was going well, although a quick preliminary test showed that the wires I was originally using to suspend the pipe did not have enough grip. After the initial construction of the pipes and wire suspenders I took one of our prototypes,we had three, home and created the first iteration of sprinklers.

- Testing (elements H-I)

Element H was well created, at least in the sense that these requirements were to ones I referenced for the rest of the project. In element I, most of my testing time was spent on sprinklers as the other parts of the prototype worked fine. During this process, despite many attempts at different shapes and sizes, and them working fine on their own, I was not able to get any of the prototype sprinklers to work.

- Evaluation (element J)

The process of getting the presentation together for the evaluation was very difficult as due to a lack of teamwork I was left to do the whole thing in class the day of the presentation. However even with that hurdle, after seeing the presentation and project most of the stakeholders had no comments and believed that the project would be successful in the future or didn't give feedback at all.

Lessons Learned About the Engineering Design Process *(K3)*

I believe that the biggest thing that should have received more attention was the researching and ideation phase. Although short cutting through researching the situation caused no problems at the beginning, during the prototyping and interaction phase I would run into problems with the sprinklers not working as expected or having to

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constantly make assumptions and last minute because I lacked the knowledge to make good decisions, which then led to even more problems. I think that I should have gone back to researching when I realized that things were not working instead of continuing to brute force it until I was too exhausted to continue. When prototyping and testing, I learned that making the prototype and requirements as accurate and specific to what you really need was very important to making sure that each part actually works how you want it to in the final. I think that in general it was also really important to acknowledge that failure was going to be inevitable and that somethings were going to be hard, as having that mindset helped me power through the boring and demotivating parts.

Element K: Reflection on the Design Project						
K1. Clear and comprehensive summary. The project designer provides __________ of the project.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	a robust and comprehensive summary of each step	a clear and comprehensive summary of the major steps	a generally clear and adequately-developed summary of most steps	a sparse summary of some steps	an overly-general summary	no summary
K2. Value judgment. A value judgment is __________ provided for project steps.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	consistently and robustly	consistently (for major steps)	consistently (for major steps, with one or two addressed in only a cursory manner)	partially (one or two addressed in only a cursory manner)	partially (and overly general or superficial)	not attempted
K3. Inclusion of lessons learned. Lessons learned	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

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reflections are __________.	substantive and are clearly useful to others attempting engineering design in the future	summarized and are clearly useful to others attempting engineering design in the future	summarized and at least most are useful to others attempting engineering design in the future	included in part and some are useful to others attempting engineering design in the future	minimally included and at least one is useful to others attempting engineering design in the future	not attempted, unclear and/or unlikely to be useful to others attempting engineering design in the future (or there is no evidence of lessons learned)